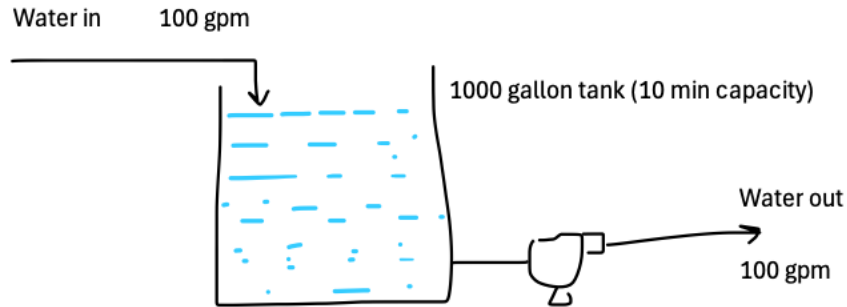


Tank and Pump system

Process Design Criteria – Base Case Report



1. System Description & Process Basis

The system is a clean-water storage and pumping arrangement. Water flows in at 100 gpm into a stainless-steel storage tank and the pump discharges 100 gpm from the tank to downstream process/piping. The tank is designed to hold approximately 10 minutes' storage at 100 gpm flow, i.e., 1,000 gallons nominal at 80% level. The tank is maintained at 80% level by a level transmitter and a feed control valve on the inlet line. The tank is covered (but atmospheric), stainless steel construction, no insulation or heat tracing. All piping is stainless steel. The pump is sized at 10 hp based on the total dynamic head (TDH) for the downstream discharge. The system is installed in Charlotte (Mecklenburg County), North Carolina (USA).

This document sets out the base case numbers, design criteria, material basis, instrumentation, controls, site environmental loads and applicable codes/standards.

2. Major Equipment & Process Data

Item	Description / Specification	Basis / Notes
Inlet flow	100 gpm (nominal)	Clean water feed to tank
Outlet flow	100 gpm (nominal)	Pump discharge rate equals inlet, steady state
Storage tank capacity (nominal)	1,000 gal @ 80% full	$100 \text{ gpm} \times 10 \text{ min} = 1,000 \text{ gal}$; design at 80% full level
Tank percent full for control	80% level maintained	Level transmitter + control feed valve logic
Tank material	Stainless steel	Corrosion-resistant for clean water service
Piping material	Stainless steel	Matches tank, avoids contamination/corrosion
Pump motor size	10 hp	Based on calculated TDH for downstream system (to be validated)
Tank pressure / atmosphere	Covered atmospheric; no special pressurization	Venting, overflow line provided
Tank insulation / heat trace	None	Service is clean ambient water – no freeze/heat control specified

Safety / access

- Tank manway, handrails or access stairs if required.
- Overflow line routed to safe location.
- Drain nozzle with isolation valve.
- Vent nozzle sized appropriately.
- Signage and labeling per site practices.

4. Site Geographic & Environmental Data – Charlotte, NC

Parameter	Value / Basis	Notes & Source
Location elevation	~ 760 ft (~ 232 m) above sea level	Charlotte elevation ~ 760 ft
Ambient temperature (design)	Min winter ambient ~ 20 °F (–6.7 °C), Max summer ambient ~ 100 °F (37.8 °C)	Use typical Charlotte data; detailed climate data to be referenced.
Relative humidity (typical)	High humidity, especially summer (~70-90 %)	Important for corrosion and equipment ventilation.
Atmospheric pressure	~ 14.7 psia at sea level; at 760 ft maybe ~14.5 psia	For tank vent/pressure design.
Wind load	To be designed per ASCE 7-16 and 2018 NCBC (ex: basic wind speed ~90 mph for Risk Category II) OSFM+1	Structural supports for tank and piping should consider wind uplift/lateral loads.
Snow load	Ground snow load in Charlotte ~ 0 psf (non-snow region) or minimal (e.g., <10 psf) per NCBC Chapter 16 ICC Digital Codes+1	Snow loads are minimal; still check local code.
Seismic load	Seismic design category likely A or B (low seismic hazard) per ASCE 7	Evaluate site-specific; Charlotte region is relatively low seismic.
Flood / flood zone	Verify project site FEMA flood zone; assume not in coastal high hazard zone	Must check actual site.
Design life	Typical mechanical / piping life expectancy ≥ 20-30 years, with corrosion allowance and maintenance.	

Codes & standards applicable for environmental/structural loads:

- 2018 North Carolina Building Code (NCBC) – Chapter 16 Structural Design for loads (snow, wind, seismic). [ICC Digital Codes+1](#)
- ASCE 7 (Minimum Design Loads for Buildings and Other Structures) for wind, snow, seismic.
- Local jurisdiction (Charlotte/Mecklenburg) may adopt additional criteria.

5. Codes, Standards & References

Below is a non-exhaustive list of codes/standards to be used in detailed engineering design of this system.

Discipline	Reference Standard	Remarks
Structural Loads	ASCE 7-16 (or latest) “Minimum Design Loads for Buildings and Other Structures”	Provides wind, snow, seismic loadings.
Building/Structural Code	2018 North Carolina Building Code (NCBC) – especially Chapter 16 “Structural Design”	Governs structural design loads in NC. ICC Digital Codes+1

Discipline	Reference Standard	Remarks
Process Piping	ASME B31.3 “Process Piping”	For stainless steel piping design, supports, stress analysis.
Pressure Vessels / Tanks	ASME Boiler & Pressure Vessel Code (BPVC) Section VIII, Div. 1 (if considered a pressure vessel)	If tank is atmospheric, may not require full pressure-vessel code; verify local jurisdiction.
Storage Tanks	API 650 “Welded Steel Tanks for Oil Storage” (for atmospheric tanks) or other appropriate standard (e.g., AWWA D100 for water tanks)	For design of stainless steel storage tank; adapt to clean water service.
Materials	ASTM A240 (for stainless steel sheet/plate), ASTM A312 (stainless piping)	Material standardization.
Pumps	Hydraulic Institute Standards (e.g., ANSI/HI standards) for pump design, testing	Ensures pump performance and selection.
Electrical / Motor	NFPA 70 (National Electrical Code), IEEE 841 (for motor if severe duty)	For motor and electrical installation.
Instrumentation/Control	ISA standards (e.g., ISA-5.1 Instrumentation Symbols), IEC 61511 (for safety instrumented systems if applicable)	For instrumentation and control system design.
Corrosion / Water Quality	ASTM, AWWA standards for water quality, corrosion control	Ensure compatibility of clean water service.
Tank Supports / Foundation	AISC Steel Construction Manual (if supports steel frames)	Structural design of supports.

6. Process Design Basis – Summary Table

Here is a consolidated table of key design basis parameters for the system:

Parameter	Value	Notes / Comments
Feed flow rate	100 gpm	Nominal steady state.
Discharge flow rate	100 gpm	Matches feed in steady state.
Tank storage time	10 minutes	Basis for storage size.
Tank nominal volume	1,000 gallons	100 gpm × 10 min.
Tank operational fill level	80 %	Controlled level of tank.
Tank material	Stainless steel (specify grade)	Suitable for clean water.
Tank design full volume required	~1,000 gallons ÷ 0.8 = ~1,250 gallons gross theoretical (to allow sump or headspace)	Additional headspace should be included; assume 1,250 gal gross capacity for design.
Tank operating pressure	Atmospheric	Venting to atmosphere; minimal internal pressure.
Tank cover	Yes (covered)	Minimizes contamination and evaporation.
Tank insulation / heat trace	None	Acceptable only if ambient conditions permit; check freezing risk.
Suction and discharge piping material	Stainless steel	Full system compatibility.
Pump motor size	10 hp	Base case; actual sizing to confirm with hydraulic head, pump curves.
Controls – Level	Level transmitter controlling inlet feed valve	Maintain ~80% level.
Controls – Inlet Valve	Automated control valve	Feed water control.
Controls – Overflow protection	Passive overflow line (and high level alarm)	Safety measure.